



Healthcare Analytics Dashboard

Project Overview

This Healthcare Analytics Dashboard was developed to analyze patient trends, hospital operations, department performance, patient satisfaction, and admission patterns.

The project demonstrates the complete Data Analytics workflow, including:

- Data Cleaning in Excel
 - Data Analysis using SQL
 - KPI Creation using DAX
 - Interactive Dashboard Development in Power BI
-



Project Objectives

The main objectives of this project were:

- Analyze patient admission trends over time
 - Monitor patient wait times
 - Evaluate patient satisfaction across departments
 - Identify high-volume departments
 - Compare patient distribution by gender
 - Build an interactive dashboard for healthcare insights
-



Tools & Technologies Used

Tool	Purpose
------	---------

Excel	Data Cleaning & Preprocessing
-------	-------------------------------

MySQL	Data Analysis & Querying
-------	--------------------------

Power BI	Dashboard Development
----------	-----------------------

Tool	Purpose
DAX	KPI & Measure Creation

Data Cleaning Process

The dataset was cleaned and prepared using Excel and SQL:

- Handled missing values
 - Standardized date formats
 - Standardized time formats
 - Corrected data types
 - Removed inconsistencies
 - Prepared dataset for reporting and visualization
-

Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

Total Patients

Tracks the total number of patients recorded.

Total Admissions

Tracks the total admitted patients.

Average Wait Time

Measures average patient waiting time.

Average Satisfaction Score

Measures overall patient satisfaction.

Dashboard Visuals

Monthly Patient Trends

Analyzes patient volume across different months.

Patients by Department

Shows departments with the highest patient load.

Gender Distribution

Compares male and female patient counts.

Average Satisfaction by Department

Highlights patient satisfaction performance across departments.

Patient Details Table

Provides detailed patient-level information for analysis.

Key Insights

1. Patient Volume Trends

Patient visits increased significantly during mid-year months and declined toward the end of the year.

2. Department Workload

No Referral and General Practice departments handled the highest number of patients.

3. Patient Satisfaction

Most departments maintained high satisfaction scores, indicating positive patient experiences.

4. Gender Distribution

Patient distribution between male and female patients was relatively balanced.

5. Operational Efficiency

Average wait time was approximately 35 minutes, indicating opportunities for workflow optimization.

Recommendations

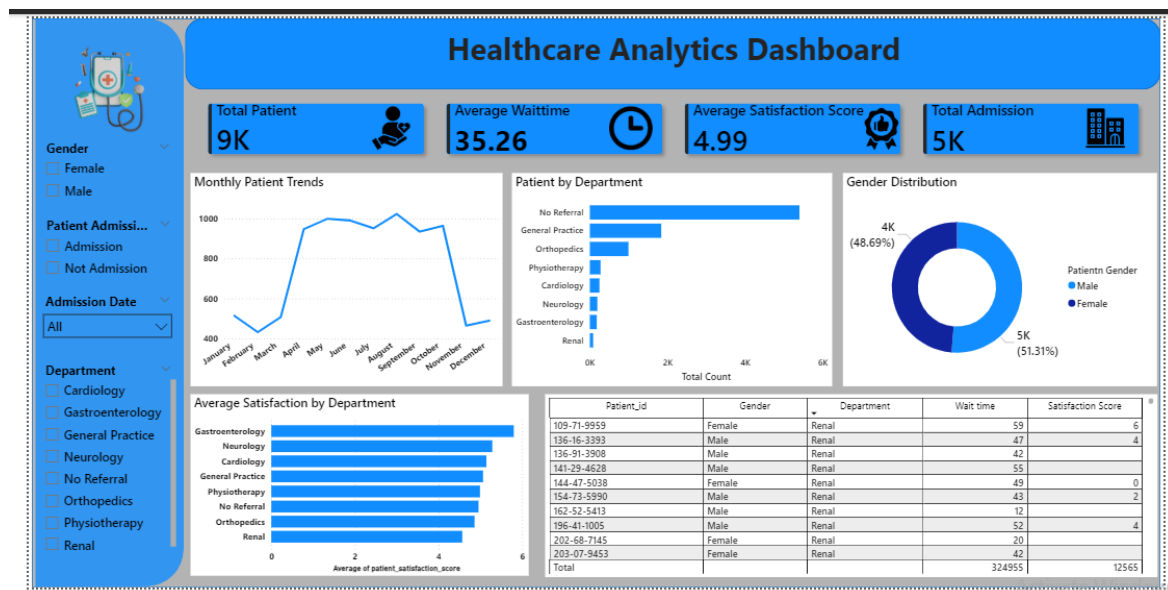
- Increase staffing during peak patient months.

- Monitor high-load departments regularly.
- Reduce patient waiting times through operational improvements.
- Continue tracking satisfaction metrics to improve patient experience.

Skills Demonstrated

- Data Cleaning
- Data Transformation
- SQL Querying
- Data Analysis
- DAX Measures
- KPI Development
- Dashboard Design
- Business Insights Generation
- Data Visualization

Dashboard Preview





Author

Himanshu Kashyap

Data Analytics Project | Excel | SQL | Power BI | DAX